25901449 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Value Addition Course (VAC) - MG4VACPHY200 - ENVIRONMENTAL PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer any **25** questions

Each question carries **2** marks

1. Which of the following rivers in India is associated with an inter-state water dispute? [K] / [CO1]

a). Narmada River

b). Godavari River

c). Mahanadi River

d). Cauvery River

2. In the nuclear fuel cycle, yellow cake refers to [U] / [CO2]

a). A type of nuclear waste stored in landfills

b). A material containing 70 to 90% Uranium oxide produced during milling

c). The powder used to compact fuel pellets

d). A byproduct of the enrichment process

3. Doubling time of population refers to: [U] / [CO3]

a). Time taken to reduce population by half

b). Time required for population to double

c). Age at which population stabilizes

d). Time taken for economic growth

4. While observing an aquatic ecosystem such as a pond, which factor was most useful in assessing habitat quality? [U] / [CO4]

a). Colour of surrounding soil

b). Water clarity and oxygen availability

c). Wind speed at the site

d). Type of fencing around the area

5. What is an important aspect of sustainable living related to species protection [U] / [CO6]

a). Replacing natural habitats with urban areas

b). Permitting the destruction of forests

c). Creating a strong public opinion to protect National Parks and Wildlife Sanctuaries

d). Not informing authorities about illegally killed species

6. Which of the following human activities contributes significantly to soil erosion? [U] / [CO1]

a). Deforestation b). Crop rotation

c). Terrace farming d). Using compost

7. Different types of plastics are difficult to recycle together because [U] / [CO2]

a). They all have same chemical makeup

b). Each type uses different polymer resins and has its own chemical make up

c). Elastics can only be recycled once before they become toxic

d). They are too heavy to transport to recycling centres

8. Narmada Bachao Andolan is associated with opposition to: [K] / [CO3]
   a). Mining projects in Odisha
   b). Industrial pollution in Delhi
   c). Large dams on the Narmada River
   d). Deforestation in the Western Ghats

9. Which of the following method can be implemented by farmers to reduce over utilization of water? [U] / [CO1]
   a). Step farming b). Organic farming
   c). Drip irrigation d). Flood irrigation

10. Geothermal energy originates from [K] / [CO1]
    a). sun
    b). wind
    c). magma inside Earth
    d). oceans

11. What is the main reason for the degradation of the grassland ecosystem? [K] / [CO1]
    a). Overgrazing
    b). Rainfall
    c). Landslides
    d). None of these

12. Amendment to Wildlife protection act in 2002 included protection of ---------- in animals group [K] / [CO2]
    a). Birds

b). Plants

c). Fishes

d). None of the above

13. The primary significance of the environmental calendar is to: [U] / [CO6]

a). Regulate industrial activities

b). Fix international environmental laws

c). Promote environmental awareness

d). Measure levels of environmental pollution

14. What is the full form of JFM? [K] / [CO1]

a). Joint Food Management b). Joint Forest Management

c). Joint Farm Management d). Joint Fertilizer management

15. What is the primary source of energy that powers all food chains in an ecosystem? [U] / [CO1]

a). Atmospheric oxygen

b). The decomposition of organic matter

c). Inorganic nutrients in the soil

d). Sun

16. Which type of terrestrial ecosystem is considered the most sensitive? [U] / [CO1]

a). Forest

b). Grassland

c). Desert

d). None of the above

17. The Montreal Protocol primarily deals with [K] / [CO2]

a). Climate change

b). Ozone layer protection

c). Biodiversity loss

d). Marine pollution

18. Bringing back an ethic for nature conservation requires: [U] / [CO6]

a). Industrial expansion

b). Strict punishment laws

c). Environmental education and conservation awareness

d). Urban development

19. Below which sound level does hearing loss generally not occur? [U] / [CO2]

a). 200 dBA

b). 80 dBA

c). 150 dBA

d). 120 dBA

20. Match the following environmental legislations (List A) with their year of enactment (List B): [K] / [CO2]

List A-Legislation

a) Air (Prevention and Control of Pollution) Act

b) Water (Prevention and Control of Pollution) Act

c)Environment (Protection) Act

List B-Year of enactment

1) 1974

2) 1986

3) 1981

a). a–3, b–1, c–2

b). a–1, b–3, c–2

c). a–2, b–1, c–3

d). a–3, b–2, c–1

21. Burning fossil fuels releases which of the following pollutants? [K] / [CO1]

a). Carbon dioxide

b). Oxides of sulphur and nitrogen

c). Carbon monoxide

d). All of the above

22. What material is added to the soil during the "liming" process? [U] / [CO6]

a). Sulphur

b). Powdered limestone

c). Aluminium

d). Chlorine

23. Vector-borne diseases increase after floods because [U] / [CO5]

a). water becomes saline

b). mosquitoes lose habitat

c). stagnant water supports breeding

d). temperature falls sharply

24. **Assertion (A):** Global warming is accelerating faster than earlier predictions. [U] / [CO6]
Reason (R): Greenhouse gas emissions continue to rise due to human activities.

a). Both A and R are true and R is the correct explanation of A

b). Both A and R are true but R is not the correct explanation of A

c). A is true but R is false

d). A is false but R is true

25. Which situation best increases the chances of human–wildlife conflict? [K] / [CO3]

a). Dense forests with no human activity

b). Buffer zones managed with community participation

c). Fragmented forests surrounded by farms and villages

d). Strictly protected core areas with no disturbance

26. Long-term solutions to reduce clashes between wildlife and people should focus on: [K] / [CO3]

a). Removing all wildlife from villages

b). Strengthening cooperation with local residents

c). Expanding urban areas into forests

d). Promoting commercial forest use

27. Which of the following was one of the first synthetic organic insecticides and was once thought to be the perfect insecticide? [K] / [CO2]

a). Malathion

b). BHC

c). Aldrin

d). DDT

28. The Bishnoi community sacrifices personal comfort mainly to protect [K] / [CO5]

a). Technology b). Nature

c). Industry d). Transport

29. Pollutants emitted directly from identifiable sources are called: [U] / [CO2]

a). Primary pollutants

b). Persistent pollutants

c). Secondary pollutants

d). Non-degradable pollutants

30. What is the term for the amount of oxygen required by bacteria to break down organic matter in water? [K] / [CO2]

a). Chemical Oxygen Requirement

b). Biological Oxygen Demand (BOD)

c). Anaerobic Respiration Index

d). Eutrophication Level

[2x25 = 50]

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